

ALFAA13443

## Potassium nitrate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 硝酸钾, ACS 最低  
**Potassium nitrate**

**Cat No. :** 13443  
**Synonyms** Saltpeter.; Nitric acid potassium salt; Niter  
**CAS No** 7757-79-1  
**Molecular Formula** K N O<sub>3</sub>

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
White

**Odor**  
Odorless

#### Emergency Overview

May intensify fire; oxidizer. Suspected of damaging fertility or the unborn child. Causes damage to organs. Causes damage to organs through prolonged or repeated exposure. May be harmful if swallowed.

#### Classification of the substance or mixture

|                                                      |            |
|------------------------------------------------------|------------|
| Oxidizing solids                                     | Category 3 |
| Acute Oral Toxicity                                  | Category 5 |
| Reproductive Toxicity                                | Category 2 |
| Specific target organ toxicity - (single exposure)   | Category 1 |
| Specific target organ toxicity - (repeated exposure) | Category 1 |

#### Label Elements



## Potassium nitrate

## Signal Word

Danger

## Hazard Statements

H272 - May intensify fire; oxidizer

H303 - May be harmful if swallowed

## Precautionary Statements

## Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P220 - Keep away from clothing and other combustible materials

P221 - Take any precaution to avoid mixing with combustibles

P270 - Do not eat, drink or smoke when using this product

P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

## Storage

P403 - Store in a well-ventilated place

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

Oxidizing. Contact with combustible material may cause fire.

## Health Hazards

Suspected of damaging fertility or the unborn child. Causes damage to organs. Causes damage to organs through prolonged or repeated exposure. May be harmful if swallowed.

## Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Potassium nitrate | 7757-79-1 | >95      |

## SECTION 4. FIRST AID MEASURES

## General Advice

If symptoms persist, call a physician.

## Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

## Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

## Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

## Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

## Most important symptoms and effects

None reasonably foreseeable.

## Self-Protection of the First Aider

No special precautions required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.). Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Sweep up and shovel into suitable containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Keep away from clothing and other combustible materials.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and

## Potassium nitrate

inhalable dust

## Exposure Controls

## Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. .

## Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard    | Glove comments                                                                 |
|----------------|-------------------|-----------------|----------------|--------------------------------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.5 mm          | EN 374 Level 6 | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Nitrile rubber |                   |                 |                |                                                                                |
| Neoprene       |                   |                 |                |                                                                                |
| PVC            |                   |                 |                |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Maintain adequate ventilation No personal respiratory protective equipment normally required

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                 |                          |                                          |
|---------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>               | White                    |                                          |
| <b>Physical State</b>           | Solid                    |                                          |
| <b>Odor</b>                     | Odorless                 |                                          |
| <b>Odor Threshold</b>           | No data available        |                                          |
| <b>pH</b>                       | 6-8                      | 5% aq. solution                          |
| <b>Melting Point/Range</b>      | 334 °C / 633.2 °F        |                                          |
| <b>Softening Point</b>          | No data available        |                                          |
| <b>Boiling Point/Range</b>      | 400 °C / 752 °F          | @ 760 mmHg                               |
| <b>Flash Point</b>              | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b> | Not flammable            |                                          |
| <b>Explosion Limits</b>         | No data available        |                                          |

**SAFETY DATA SHEET****Potassium nitrate**

|                                                |                          |                      |
|------------------------------------------------|--------------------------|----------------------|
| <b>Vapor Pressure</b>                          | No data available        |                      |
| <b>Vapor Density</b>                           | Not applicable           | Solid                |
| <b>Specific Gravity / Density</b>              | 2.1 @ 20 °C              | Literature reference |
| <b>Bulk Density</b>                            | No data available        |                      |
| <b>Water Solubility</b>                        | Soluble                  |                      |
| <b>Solubility in other solvents</b>            | No information available |                      |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                      |
| <b>Autoignition Temperature</b>                | No data available        |                      |
| <b>Decomposition Temperature</b>               | > 400°C                  |                      |
| <b>Viscosity</b>                               | Not applicable           | Solid                |
| <b>Explosive Properties</b>                    | No information available |                      |
| <b>Oxidizing Properties</b>                    | Oxidizer                 |                      |

|                          |        |
|--------------------------|--------|
| <b>Molecular Formula</b> | K N O3 |
| <b>Molecular Weight</b>  | 101.1  |

**SECTION 10. STABILITY AND REACTIVITY**

|                                 |                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------|
| <b>Stability</b>                | Oxidizer: Contact with combustible/organic material may cause fire.             |
| <b>Hazardous Reactions</b>      | None under normal processing.                                                   |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.                                        |
| <b>Conditions to Avoid</b>      | Avoid dust formation. Excess heat. Combustible material. Incompatible products. |
| <b>Materials to avoid</b>       | Strong reducing agents. Strong acids. Combustible material.                     |

**Hazardous Decomposition Products** Potassium oxides. Nitrogen oxides (NOx).

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| Component         | LD50 Oral                 | LD50 Dermal        | LC50 Inhalation      |
|-------------------|---------------------------|--------------------|----------------------|
| Potassium nitrate | LD50 = 3015 mg/kg ( Rat ) | > 5000 mg/kg (Rat) | >0.527 mg/l 4h (Rat) |

**(b) skin corrosion/irritation;** Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;** Based on available data, the classification criteria are not met  
**Test method** OECD 405  
**Test species** rabbit  
**Observation end point** No eye irritation

**(d) respiratory or skin sensitization;**  
**Respiratory** Based on available data, the classification criteria are not met  
**Skin** Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met  
Did not show mutagenic effects in animal experiments

**(f) carcinogenicity;** Based on available data, the classification criteria are not met  
The table below indicates whether each agency has listed any ingredient as a carcinogen

## Potassium nitrate

(g) reproductive toxicity;  
Reproductive Effects Based on available data, the classification criteria are not met  
Animal testing did not show any effects on fertility.

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Test method OECD Test Guideline 422  
Study result NOAEL = 1500 mg/kg bw/day  
Target Organs None known.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects This product contains the following substance(s) which are hazardous for the environment. .

| Component         | Freshwater Fish      | Water Flea          | Freshwater Algae          | Microtox |
|-------------------|----------------------|---------------------|---------------------------|----------|
| Potassium nitrate | 1378 mg/l LC50 (96h) | 490 mg/l EC50 (48h) | > 1700 mg/l EC50 (10 day) |          |

## Persistence and Degradability

Persistence Soluble in water, Persistence is unlikely, based on information available.  
Degradability Not relevant for inorganic substances.

Bioaccumulative Potential Bioaccumulation is unlikely

Mobility in soil The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors  
Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

Other Information Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## SECTION 14. TRANSPORT INFORMATION

## Road and Rail Transport

## SAFETY DATA SHEET

## Potassium nitrate

UN-No UN1486  
Proper Shipping Name POTASSIUM NITRATE  
Hazard Class 5.1  
Packing Group III

IMDG/IMO

UN-No UN1486  
Proper Shipping Name POTASSIUM NITRATE  
Hazard Class 5.1  
Packing Group III

IATA

UN-No UN1486  
Proper Shipping Name POTASSIUM NITRATE  
Hazard Class 5.1  
Packing Group III

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component         | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Potassium nitrate | X                                                   | X                                       | X    | X     | 231-818-8 | X    | X   | X     | X    | X    | X    | KE-29163 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
Creation Date 05-Aug-2010  
Revision Date 07-Mar-2024  
Revision Summary New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

CAS - Chemical Abstracts Service

TSCA - United States Toxic Substances Control Act Section 8(b)  
Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical

## Potassium nitrate

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Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances  
**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**